Vectors I Homework 3: Vector Product — Solutions

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Official worked solutions for sourced past-paper questions; True/False answers from the Basics Worksheet.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$. $\lvert\mathbf{a}\times\mathbf{a}\rvert=\lvert\mathbf{a}\rvert^2\sin 0^\circ=0$	**T**
(b)	$\mathbf{a}\times\mathbf{a}=0$. $\mathbf{a}\times\mathbf{a}=\mathbf{0}\neq 0$ (left side is the zero vector)	**F**
(c)	$\mathbf{i}\times\mathbf{j}=\mathbf{k}$. $\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}$	**T**
(d)	$\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k},\ \mathbf{j}\times\mathbf{i}=-\mathbf{k}$	**F**
(e)	$\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$. $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$	**T**
(f)	$\mathbf{a}\times\mathbf{b}$ is a scalar. $\mathbf{i}\times\mathbf{j}=\mathbf{k}$ (a vector)	**F**
(g)	$\mathbf{k}\times\mathbf{j}=\mathbf{i}$. $\mathbf{j}\times\mathbf{k}=\mathbf{i}\ \therefore\ \mathbf{k}\times\mathbf{j}=-\mathbf{i}$	**F**
(h)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$. $\begin{pmatrix}2\\4\\6\end{pmatrix}=2\begin{pmatrix}1\\2\\3\end{pmatrix}$	**T**
(i)	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$. Distributive law	**T**
(j)	$(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$. $(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})=\lambda(\mathbf{a}\times\mathbf{b})$	**T**
(k)	$\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0,\ (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$	**T**
(l)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\text{vector}\cdot\text{vector}$	**T**
(m)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Definition of vector product	**T**

	Statement	**T/F**
(n)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\times\mathbf{i}\rvert=0\neq 1$	**F**
(o)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\times 2\mathbf{i}=\mathbf{0},\ \mathbf{i}\neq\mathbf{0},\ 2\mathbf{i}\neq\mathbf{0}$	**F**
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$. $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=0\ \therefore\ \theta=0^\circ$ or 180°	**T**
(q)	The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $\lvert\mathbf{a}\times\mathbf{b}\rvert$. Area $=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=\lvert\mathbf{a}\times\mathbf{b}\rvert$	**T**
(r)	The area of triangle OAB is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$. Area $=\frac{1}{2}\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$	**F**
(s)	$\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore$ meaningless	**F**
(t)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$. $\sin\theta=1\ \therefore\ \theta=90^\circ$	**T**
(u)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$. both $=\det(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$	**T**
(v)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2(\sin^2\theta+\cos^2\theta)=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$	**T**
(w)	$\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{i}\times\mathbf{j}=\mathbf{k}=\mathbf{i}\times(\mathbf{i}+\mathbf{j}),\ \mathbf{j}\neq\mathbf{i}+\mathbf{j}$	**F**
(x)	$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$. $\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j},\ (\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}$	**F**
(y)	The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$. distance $=\lvert\overrightarrow{AB}\times\hat{\mathbf{d}}\rvert=\dfrac{\lvert\overrightarrow{AB}\times\mathbf{d}\rvert}{\lvert\mathbf{d}\rvert}$	**F**
(z)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2=\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=0$	**T**

2 [ASRJC 2025 Prelim P2 Q1]

Official solution.

Since $AM:AB=3:4$, we have $AM:MB=3:1$. By the ratio theorem,

$$\overrightarrow{OM}=\frac{\mathbf{a}+3\mathbf{b}}{4}.$$

$$\text{Area of } \Delta OAM=\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OM}\right|=\frac{1}{2}\left|\mathbf{a}\times\frac{\mathbf{a}+3\mathbf{b}}{4}\right|=\frac{1}{8}\left|\mathbf{a}\times\mathbf{a}+3\mathbf{a}\times\mathbf{b}\right|=\frac{3}{8}\left|\mathbf{a}\times\mathbf{b}\right|$$
$$=\frac{3}{8}|\mathbf{a}||\mathbf{b}|\sin\frac{\pi}{6}=\frac{3}{8}\cdot k\cdot 1\cdot\frac{1}{2}=\frac{3k}{16}\text{ units}^2.$$

3 [EJC 2025 Promo Q3]

Official solution.

(a)

We can deduce that vectors **a** and **b** are parallel, or either vector **a** or **b** is the zero vector.

(b)

$$\mathbf{n}=\frac{1}{\sqrt{14}}\begin{pmatrix}2\\-1\\3\end{pmatrix} \quad \text{or} \quad \mathbf{n}=-\frac{1}{\sqrt{14}}\begin{pmatrix}2\\-1\\3\end{pmatrix}.$$

(c)

Let θ be the acute angle between $2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and the y-axis.

$$\cos\theta=\frac{\left|\begin{pmatrix}2\\-1\\3\end{pmatrix}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}\right|}{\sqrt{14}\cdot 1}=\frac{1}{\sqrt{14}}.$$

4 [NYJC 2025 Prelim P1 Q3]

Official solution.

(a)

$AB^2+BC^2=AC^2 \Rightarrow |\overrightarrow{AB}|^2+|\overrightarrow{BC}|^2=|\overrightarrow{AC}|^2$. Since $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$,

$$\begin{aligned}(\overrightarrow{AB}+\overrightarrow{BC})\cdot(\overrightarrow{AB}+\overrightarrow{BC})&=\overrightarrow{AC}\cdot\overrightarrow{AC}\\ |\overrightarrow{AB}|^2+2\overrightarrow{AB}\cdot\overrightarrow{BC}+|\overrightarrow{BC}|^2&=|\overrightarrow{AC}|^2\\ |\overrightarrow{AC}|^2+2\overrightarrow{AB}\cdot\overrightarrow{BC}&=|\overrightarrow{AC}|^2 \quad (\text{using } AB^2+BC^2=AC^2)\\ \overrightarrow{AB}\cdot\overrightarrow{BC}&=0.\end{aligned}$$

So $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{BC}$ and hence $\angle ABC=90^\circ$.

(b)

By the ratio theorem, $\mathbf{d}=(1-\lambda)\mathbf{a}+\lambda\mathbf{c}$.

$$\begin{aligned}\text{Area of } \Delta ABD&=\frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{BD}\right|=\frac{1}{2}\left|(\mathbf{b}-\mathbf{a})\times(\mathbf{d}-\mathbf{b})\right|\\ &=\frac{1}{2}\left|(\mathbf{b}-\mathbf{a})\times((1-\lambda)\mathbf{a}+\lambda\mathbf{c}-\mathbf{b})\right|\\ &=\frac{\lambda}{2}\left|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}\right|.\end{aligned}$$

Hence $k=\frac{\lambda}{2}$.

5 [NJC 2025 Prelim P2 Q3]

Official solution.

(i)

$$\mathbf{p}=\begin{pmatrix}\cos t\\ \sin t\\ -1\end{pmatrix}, \qquad \mathbf{q}=\begin{pmatrix}\cos 2t\\ -\sin 2t\\ \alpha\end{pmatrix}.$$

For O, P, Q collinear, $\mathbf{p}\times\mathbf{q}=\mathbf{0}$. Equating the **k**-component to 0:

$$-\sin 2t\cos t-\cos 2t\sin t=0$$

$$-\sin 3t=0 \qquad \Rightarrow \qquad t=\frac{\pi}{3} \text{ or } \frac{2\pi}{3} \quad (0<t<\pi).$$

When $t=\dfrac{\pi}{3}$: $-(\alpha\cos\frac{\pi}{3}+\cos\frac{2\pi}{3})=0\Rightarrow\alpha=1$.

When $t=\dfrac{2\pi}{3}$: $-(\alpha\cos\frac{2\pi}{3}+\cos\frac{4\pi}{3})=0\Rightarrow\alpha=-1$.

Final answers: $t=\dfrac{\pi}{3}$, $\alpha=1$; $t=\dfrac{2\pi}{3}$, $\alpha=-1$.

(ii)

$$\mathbf{p}\cdot\mathbf{q}=\cos t\cos 2t-\sin t\sin 2t-0.5=\cos 3t-0.5 \quad \text{(shown)}.$$

Length of projection of $\mathbf{q}$ onto $\mathbf{p}=\dfrac{|\mathbf{p}\cdot\mathbf{q}|}{|\mathbf{p}|}=\dfrac{|\cos 3t-0.5|}{\sqrt{2}}$.

Greatest length when $\cos 3t=-1$:

$$\frac{|-1-0.5|}{\sqrt{2}}=\frac{1.5}{\sqrt{2}}=\frac{3}{2\sqrt{2}} \quad (=1.06 \text{ to 3 s.f.}).$$

6 [CJC 2025 Promo Q6]

Official solution.

(a)

Since AB is a diameter through O, $\mathbf{b}=-\mathbf{a}$.

$$\overrightarrow{AC}=\mathbf{c}-\mathbf{a}, \qquad \overrightarrow{BC}=\mathbf{c}-\mathbf{b}=\mathbf{c}+\mathbf{a}$$

$$\overrightarrow{AC}\cdot\overrightarrow{BC}=(\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}+\mathbf{a})=|\mathbf{c}|^2-|\mathbf{a}|^2=0$$

(since $|\mathbf{c}|=|\mathbf{a}|=$ radius). Hence AC and BC are perpendicular.

(b)

$|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$ is the length of the projection of $\overrightarrow{AC}$ onto $\overrightarrow{OA}$.

(c)

By the ratio theorem, $\overrightarrow{OM}=\dfrac{\overrightarrow{OD}+\overrightarrow{OA}}{2}=\dfrac{1}{2}(\mathbf{c}+2\mathbf{a}+\mathbf{a})=\dfrac{1}{2}\mathbf{c}+\dfrac{3}{2}\mathbf{a}$ (since $\overrightarrow{OD}=\mathbf{c}+2\mathbf{a}$ with $\mathbf{b}=-\mathbf{a}$). Also $\overrightarrow{OR}=3\mathbf{a}$.

$$\begin{aligned}\overrightarrow{CM}\times\overrightarrow{CR}&=\left(\overrightarrow{OM}-\mathbf{c}\right)\times\left(\overrightarrow{OR}-\mathbf{c}\right)=\left(\tfrac{1}{2}\mathbf{c}+\tfrac{3}{2}\mathbf{a}-\mathbf{c}\right)\times(3\mathbf{a}-\mathbf{c})\\&=\left(\tfrac{3}{2}\mathbf{a}-\tfrac{1}{2}\mathbf{c}\right)\times(3\mathbf{a}-\mathbf{c})=\tfrac{9}{2}(\mathbf{a}\times\mathbf{a})-\tfrac{3}{2}(\mathbf{a}\times\mathbf{c})-\tfrac{3}{2}(\mathbf{c}\times\mathbf{a})+\tfrac{1}{2}(\mathbf{c}\times\mathbf{c})\\&=0-\tfrac{3}{2}\mathbf{a}\times\mathbf{c}+\tfrac{3}{2}\mathbf{a}\times\mathbf{c}+0=\mathbf{0}.\end{aligned}$$

Hence $\overrightarrow{CM}$ and $\overrightarrow{CR}$ are parallel. Since C is a common point, C, M and R are collinear.

Alternatively: $\overrightarrow{CM}=\frac{3}{2}\mathbf{a}-\frac{1}{2}\mathbf{c}$ and $\overrightarrow{CR}=3\mathbf{a}-\mathbf{c}=2\overrightarrow{CM}$.

END OF SOLUTIONS